

Transferring single cell tools to spatial data

Some current spatial methods and their characteristics

10x Genomics Visium

- Whole transcriptome
- Spot-based (55 μ m diameter, 100 μ m distance)
- No segmentation

10x Genomics Xenium

- Preselected probe panel
- Accurate location of transcripts
- Segmentation possible

OpenST

- Whole transcriptome
- Spot-based (0.6 μ m distance)
- Segmentation possible

Stereoseq

- Whole transcriptome
- Grid-based (0.5 μ m distance)
- Segmentation possible

Bruker CosMx

- Preselected probe panel
- Accurate location of transcripts
- Segmentation possible

10x Genomics Atera

- Preselected probe panel (bordering on whole transcriptome)
- Accurate location of transcripts
- Segmentation possible

What is segmentation ?
and
Why is it so important ?

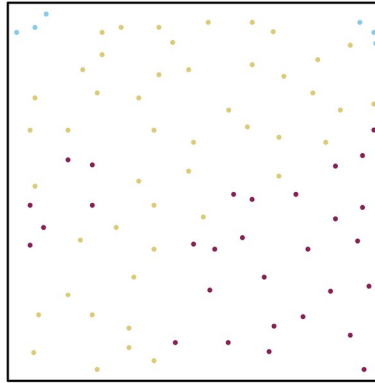
How are cell boundaries defined? (using a probe-based method as an example)

Tissue w/ probe readout

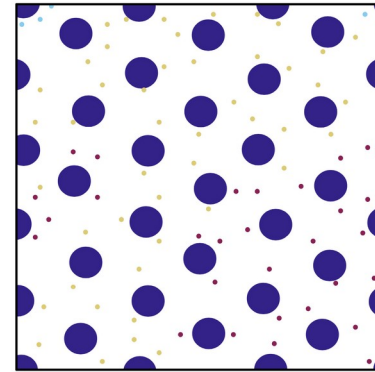


B-cell
markers
T-cell
markers
Macrophage
markers

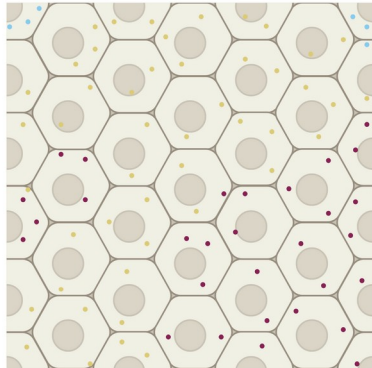
Probe readout



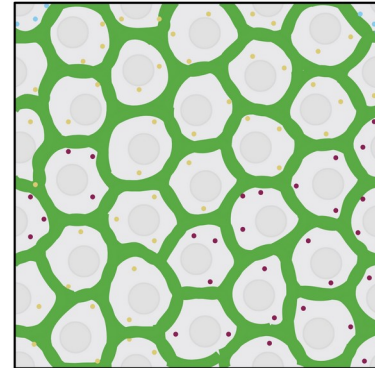
Probe readout w/ nuclear stain



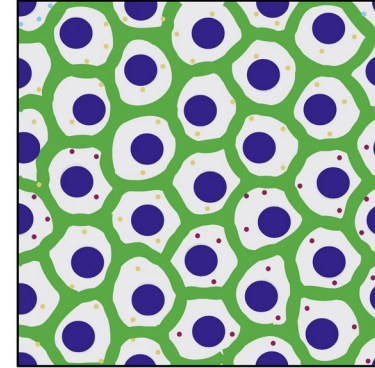
Probe readout w/ nuclear expansion



Probe readout w/
boundary stain



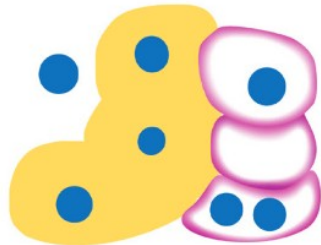
Probe readout w/ boundary stain &
nuclear stain



Boundary stain: combination of antibodies staining membrane proteins. Nuclear stain: dapi

Xenium additionally adds interior RNA stain

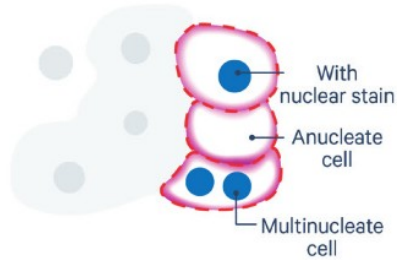
Types of Stains



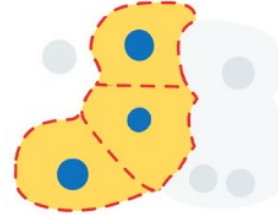
- Nuclear
- Interior
- Boundary

Types of Cell Segmentation

Boundary stain
(does not require nuclear staining)



Interior stain and expansion
(requires nuclear staining)



----- Inferred cell segmentation

Nuclear expansion
(requires nuclear staining)



*Expands 5µm or until another cell boundary is encountered

Image by 10x Genomics

Selected computational approaches to cell segmentation

Imaging-based

Cell boundaries based on staining images



DeepCell



ilastik

(+ proprietary algorithms by instrument vendors)

Transcript based

Cell boundaries based on transcript distribution & probability of transcripts occurring together



Baysor



proseg

Assuming successful segmentation...

(in Visium the spots represent a mixture of spatially close cells instead)

Can we transfer all our single cell tools right away?

Important differences in single cell and spatial analysis

- Cells are less 'independent' compared to scRNAseq due to spatial proximity / autocorrelation
 - Multiple testing (e.g. differential gene expression) becomes complicated
 - Batch effects can have a spatial component
 - For meaningful statistics, multiple biological samples need to be recorded and analysed
 - Cell type annotation can be aided by microscopy images & additional staining
 - Neighbourhood analysis should be based on actual recorded physical proximity, not inferred proximity
- Pooling from multiple animals / donors like in scRNAseq is not possible, cells/spots in one sample are never independent from each other

Selected spatial analysis tools

Python

- Squidpy

<https://squidpy.readthedocs.io/en/stable/>

- Scanpy (specific spatial functionality is increasingly moved to squidpy now)

<https://scanpy.scverse.org/en/stable/>

- Cell2 location (Visium deconvolution)

<https://github.com/BayraktarLab/cell2location>

R

- Seurat (v5+)

<https://satijalab.org/seurat/>

- Giotto

<https://giottosuite.com/index.html>

- VoltRon (especially useful for microscopy alignment and region-based analysis)

<https://bioinformatics.mdc-berlin.de/VoltRon/>